

Question:

Jen enters a lottery by picking 4 distinct numbers from $S=\{1,2,3,\cdots,9,10\}$. 4 numbers are randomly chosen from S . She wins a prize if at least two of her numbers were 2 of the randomly chosen numbers, and wins the grand prize if all four of her numbers were the randomly chosen numbers. The probability of her winning the grand prize given that she won a prize is $\frac{m}{n}$ where m and n are relatively prime positive integers. Find $m+n$.

Reasoning + Answer:

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1. Identify the total number of ways to choose 4 numbers from 10:

The total number of ways to choose 4 numbers from the set $(S = \{1, 2, 3, \ldots, 10\})$ is given by the combination formula $(\binom{10}{4})$.

$$\binom{10}{4} = \frac{10!}{4!(10-4)!} = \frac{10!}{4! \cdot 6!} = 210$$

2. Calculate the number of ways Jen can win a prize (at least two of her numbers match the randomly chosen numbers):

- The scenario can be broken down into two parts: she picks exactly 2 of the 4 chosen numbers, or exactly 3 of the 4 chosen numbers, or all 4 of the chosen numbers.

Case 1: Exactly 2 chosen numbers match:

$$\binom{4}{2} \cdot \binom{6}{2} = 6 \cdot 15 = 90$$

Case 2: Exactly 3 chosen numbers match:

$$\binom{4}{3} \cdot \binom{6}{1} = 4 \cdot 6 = 24$$

Case 3: All 4 chosen numbers match:

$$\binom{4}{4} = 1$$

Adding these together gives the number of ways Jen can win a prize:

$$90 + 24 + 1 = 115$$

3. Calculate the number of ways Jen can win the grand prize:

$$\binom{4}{4} = 1$$

4. Probability of winning the grand prize given that she won a prize:

The probability is the ratio of the number of ways to win the grand prize to the number of ways to win a prize.

$$P(\text{win grand prize} \mid \text{win a prize}) = \frac{\text{Number of ways to win the grand prize}}{\text{Number of ways to win a prize}} = \frac{1}{115}$$

5. Identify (m) and (n) :

The fraction $(\frac{1}{115})$ is already in simplest form, so $(m = 1)$ and $(n = 115)$.

6. Find $(m + n)$:

$$m + n = 1 + 115 = 116$$

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<answer> 116 </answer>